

MedTriageAI

Cloud-Based AI-Assisted Chest X-Ray Triage Platform

Technical Report

Cloud Computing · Year III, Semester II · May 2026

1. Case Study — Project in National and International Reality

1.1 The Problem We Are Solving

Chest X-ray (CXR) is the most frequently performed medical imaging exam worldwide, with over 2 billion CXRs acquired every year. Despite this volume, the supply of qualified radiologists is structurally insufficient to interpret them in a timely manner. The bottleneck is most acute in emergency departments, rural clinics, and during off-hours shifts, where a CXR may wait hours or even days for a formal read.

MedTriageAI is a multi-tenant cloud-native SaaS platform that takes a chest X-ray and returns, in under a minute: probability scores for 14 thoracic pathologies, a Grad-CAM heatmap localising the most suspicious region, and a structured plain-language explanation generated by a multimodal LLM. A clinician then reviews, edits, signs off, and the system delivers a PDF report to the patient. It is explicitly not a diagnostic device — it is a triage and prioritisation aid for non-radiologist clinicians.

1.2 International Reality

Roughly two-thirds of the world's population lacks access to basic medical imaging interpretation, according to a 2023 Lancet Digital Health commentary. In sub-Saharan Africa the radiologist density is approximately one per million inhabitants. Even in the United Kingdom, the Royal College of Radiologists reported a 30% workforce shortfall in its 2023 Census.

The global AI-in-medical-imaging market was valued at approximately USD 1.3 billion in 2023 and is projected to reach USD 8.2 billion by 2030, growing at a compound annual rate of roughly 35% (Grand View Research, April 2026). The U.S. FDA had authorized 950 AI/ML-enabled medical devices as of August 2024, with the majority in radiology. The WHO and Stop TB Partnership endorse AI-based CXR screening as a frontline tool for tuberculosis case-finding in high-burden countries.

1.3 National Reality (Romania)

Romania has approximately 5 to 7 radiologists per 100 000 inhabitants, well below the EU average of around 12.7 per 100 000 (Brady et al., EU-REST Study, Insights into Imaging, 2025). Public hospitals routinely report 1 to 3 day delays for non-emergency CXR reads.

Romania has one of the highest tuberculosis incidence rates in the EU at 55 per 100 000 in 2023, accounting for approximately 25% of all TB cases reported across EU/EEA countries (World Bank / ECDC TB Surveillance Report, 2025). Romania's PNRR commitments include EUR 535 million for eight new hospitals and EUR 100 million for the PIAS digital health platform, both financed through the EU Recovery and Resilience Facility, creating a concrete procurement window for cloud-based AI tools.

1.4 Where MedTriageAI Fits

MedTriageAI targets the gap between image acquisition and radiologist read. The primary customer segments are county emergency hospitals that lack 24/7 radiology coverage, family-medicine clinics in rural counties using portable X-ray units, NGO and screening programs where throughput matters more than full diagnostic reads, and telemedicine providers needing an automated first-pass triage. Unlike established enterprise players, MedTriageAI is designed from day one as a multi-tenant SaaS on Google Cloud Platform, with a variable cost of approximately €0.005 to €0.01 per case at published GCP list prices (May 2026), removing the capital expenditure barrier that has kept Romanian public hospitals out of the AI-radiology market.

2. Existing Solutions — Competitive and Technical Landscape

2.1 Competitor Overview

We benchmarked five solutions that overlap with MedTriageAI's scope.

Aidoc (Israel) covers multiple modalities including CT and CXR. It uses an on-prem appliance combined with an AWS cloud backbone where a DICOM router intercepts PACS traffic. It holds FDA clearance for over 15 algorithms and CE marking, and goes to market through direct enterprise subscription sales.

Lunit INSIGHT CXR (South Korea) focuses exclusively on chest X-ray and uses a hybrid architecture combining cloud inference with an on-prem PACS plug-in. It is FDA-cleared, CE-marked, and KFDA-approved, reaching the market through partnerships with GE Healthcare and Philips.

Qure.ai (India) covers CXR and head CT. It is cloud-native on AWS with REST and DICOMweb APIs and supports an offline mode for low-connectivity regions. It holds CE marking and FDA clearance, has a WHO endorsement for TB screening, and sells on a per-scan model through WHO and NGO pricing tiers.

Annalise.ai (Australia / UK) covers CXR and brain CT. It is cloud-native on Azure with a workflow-integrated viewer, holds CE marking and TGA approval, and sells through hospital enterprise contracts with viewer plug-ins for Sectra and Agfa.

Google Health (USA) has published CXR research using TPU-trained transformer models but has not commercialised a standalone product; access is via the Cloud Healthcare API on an OEM licensing basis.

2.2 Architectural Patterns Observed

All commercial competitors integrate with hospital PACS via DICOM C-STORE or DICOMweb rather than file uploads, which is why MedTriageAI includes a DICOM gateway for the enterprise tier. Aidoc and Lunit deploy an on-prem appliance for latency, privacy, and offline operation; MedTriageAI's CheXNet container is packaged so the same artifact can be deployed on Cloud Run for the SaaS tier or on-prem Kubernetes for enterprise customers. Radiologists do not switch tools: Aidoc and Annalise embed results directly in the existing PACS viewer; MedTriageAI exposes a FHIR/HL7 webhook for enterprise customers so results surface in whatever system they already use. Finally, modern systems combine specialised CNNs with downstream LLMs for narrative report generation, which is exactly the CheXNet plus Gemini pattern we adopted.

2.3 Marketing Approaches Observed

Every competitor leads with clinical evidence: peer-reviewed publications and regulatory clearances are prominently featured, and marketing is B2B and evidence-driven. Outcome-based pricing on a per-scan or per-bed basis is replacing capital-expenditure licenses. Public case studies with brand-name hospitals such as Mayo Clinic, NHS trusts, and AIIMS serve as the primary trust signal. Presence at RSNA in Chicago and ECR in Vienna is where this market is made; all major players maintain booths at both events.

2.4 Where MedTriageAI Differentiates

Competitors require an on-prem appliance and an enterprise contract negotiated over months. MedTriageAI is a pure cloud SaaS with self-service sign-up. The entry price for a competitor is a five-figure annual license; MedTriageAI targets sub-€0.80 per case on a pay-as-you-go basis. The target buyer for competitors is the hospital CIO or Head of Radiology; MedTriageAI targets the family-medicine clinic, county ED, and NGO program. LLM-generated explanation is limited or experimental in every competitor; in MedTriageAI it is native, with Gemini 2.5 Flash built directly into the pipeline. Competitors hold Class IIa medical device clearances under

MDR; MedTriageAI intentionally occupies a narrower scope as a decision-support tool rather than a diagnostic device.

3. Technologies — Overview and Motivation

The system is organised into six layers: client, edge and identity, application, asynchronous pipeline, AI/ML services, and data/platform/operations.

3.1 Client Tier

The patient and clinician UI is built with vanilla HTML, CSS, and JavaScript. Avoiding a SPA framework cuts complexity, bundle size, and build cost; files are served statically by App Engine. The Firestore Web SDK is used for realtime listeners: the result page subscribes to the case document and re-renders automatically as the pipeline transitions from pending to vision_done to complete, eliminating polling. Chart.js renders the score-bar visualisation for CheXNet pathology probabilities.

3.2 Edge and Identity

Google Cloud API Gateway handles TLS termination, OpenAPI-driven routing, per-key quotas, and request validation. The same OpenAPI document published on SwaggerHub is the source of truth for both client SDKs and the gateway configuration. Firebase Authentication manages identity for three actor classes — patient, clinician, and admin — issuing JWTs with custom claims for tenant ID and role that integrate natively with Firestore security rules. Cloud Armor provides WAF and DDoS protection in front of the gateway, meeting the compliance baseline required for any internet-facing healthcare API.

3.3 Application Tier

The API and orchestration server runs Node.js 20 with Express 5 on App Engine Standard. The ecosystem provides mature GCP SDK support and the async I/O model fits a pipeline that fans out to multiple long-running services. App Engine Standard auto-scales to zero and the F1 instance class fits the free tier. Multer handles multipart upload streaming directly to Cloud Storage without temp files. Joi and Zod provide request-body schema validation as a defense-in-depth layer on top of the API Gateway validation.

3.4 Asynchronous Pipeline

Cloud Pub/Sub decouples pipeline stages using at-least-once delivery with exponential back-off and dead-letter topics. Three topics are used: case-created, vision-done, and case-signed-off. Cloud Functions (2nd gen) hosts the per-stage workers — the Vision plus CheXNet worker, the Gemini worker, the PDF worker, and the email worker — each scaling independently per topic and billed per invocation. Cloud Tasks handles outbound webhook delivery and email enqueueing with first-class retry semantics.

3.5 AI and ML Services

CheXNet, a DenseNet-121 model peer-reviewed by Rajpurkar et al. (2017) and trained on ChestX-ray14 comprising 112 120 images, runs containerised on Cloud Run. It returns probability scores for 14 thoracic pathologies together with a Grad-CAM heatmap as a base64 PNG. Cloud Run scales to zero, keeping idle cost near zero. Cloud Vision API performs generic image labelling and object localisation as a sanity check that the uploaded file is actually an X-ray, and provides supplementary context to the LLM.

Google Gemini 2.5 Flash is the multimodal model that generates structured findings, a plain-language explanation, severity classification, and a primary diagnosis label. JSON-mode output integrates cleanly with Firestore. The design philosophy is specialised model first, generalist model second: CheXNet provides the calibrated numerical signal; Gemini provides the narrative. At published Vertex AI pricing of \$0.30 per million input tokens and \$2.50 per million output tokens (May 2026), the Gemini cost per case is approximately €0.003.

Memorystore Redis caches all AI responses keyed by the SHA-256 hash of the image bytes, so duplicate uploads do not re-spend on inference.

The DICOM gateway runs on Cloud Run using the dcm4che toolkit. It accepts DICOM C-STORE and DICOMweb from hospital PACS, anonymises PHI tags according to tenant configuration, transcodes pixel data to JPEG, and hands off to the regular upload pipeline.

3.6 Reporting and Delivery

A Cloud Run service running Puppeteer with Handlebars templates renders signed PDF reports from case and review data. Headless Chrome produces print-grade PDFs from the same HTML template used in the UI. SendGrid delivers transactional email to patients via Cloud Tasks. Patient access to the PDF is via Cloud Storage signed URLs that expire after seven days.

3.7 Data Tier

Cloud Storage holds the raw image, Grad-CAM heatmap PNG, and generated PDF report, with lifecycle rules that auto-delete raw images after 30 days. Firestore in Native mode is the source of truth for case documents, user profiles, review history, and the audit log; security rules enforce tenant and role isolation. BigQuery receives a daily de-identified export of case metadata for dashboards covering case volume, severity distribution, model agreement, and per-pathology recall, visualised through Looker Studio.

3.8 Platform, Security, and Operations

Secret Manager stores the Gemini API key, SendGrid key, and partner credentials centrally, removing secrets from app.yaml and enabling rotation without redeploy. Cloud Logging, Cloud Monitoring, and Cloud Trace provide structured logs, SLO dashboards, alerting policies, and distributed tracing across the Express server, Pub/Sub, and worker chain. Cloud Build runs the CI/CD pipeline — tests, SonarCloud static analysis, quality gate enforcement — before deploying to App Engine, Cloud Run, and Cloud Functions. Terraform provisions all GCP resources as Infrastructure-as-Code, enabling reproducible environments from a single command.

3.9 Why GCP

Gemini on Vertex AI provides a first-party multimodal LLM without leaving the cloud trust boundary, important for healthcare data residency. Cloud Run, App Engine Standard, and Cloud Functions all scale to zero, making the cost model ideal for early-stage bursty traffic. The europe-west6 region satisfies GDPR data-residency commitments to Romanian customers. Vertex AI provides a managed upgrade path to fine-tuned models without re-platforming.

4. Business Model Canvas

Key Partners

Google Cloud provides the infrastructure. Romanian county hospitals serve as pilot partners and clinical validation sites. The University of Medicine supplies clinical advisors. PACS vendors are integration partners for the DICOM gateway. CNAS and the Ministry of Health are procurement stakeholders under PNRR. Stop TB and WHO provide partnership channels for low-resource markets. SendGrid handles transactional email. SonarCloud and GitHub support code quality and version control.

Key Activities

The core activities are pipeline research and development covering CheXNet and Gemini integration, clinical validation studies to support regulatory and sales positioning, regulatory work on the CE non-diagnostic pathway, DICOM and FHIR integration engineering, customer support, and B2B and B2B2C sales.

Value Propositions

For non-radiologist clinicians: from X-ray to a structured, ranked, and explained finding in under 60 seconds, at less than €1 per case. For health systems: zero capital expenditure, GDPR-resident SaaS for AI triage that lifts throughput without a new appliance. For NGOs: AI CXR triage deployable from any browser with no installation and no PACS required.

Customer Relationships

SMB customers self-serve through the web sign-up flow. Hospital customers receive a dedicated onboarding process. Key opinion leader accounts have a clinical liaison. All customers have access to email and ticket support, and a community of practice forum serves the broader user base.

Customer Segments

The primary segments are county emergency hospitals without 24/7 radiology cover and family-medicine clinics in rural areas. Secondary segments include TB and occupational lung disease screening programs, telemedicine providers, and NGOs operating in low-resource regions.

Key Resources

The CheXNet model weights and Grad-CAM implementation, a curated CXR evaluation set for internal benchmarking, GCP credits, the engineering team, clinical advisors, and the CE non-diagnostic self-assessment documentation.

Channels

The primary channel is direct web via medtriage.ai. Conference presence at RSNA and ECR supports B2B sales. The PACS vendor marketplace provides an integration channel for enterprise. Public procurement under PNRR creates a Romanian-specific channel. Clinical publications and NGO partnerships reach the secondary segments.

Cost Structure

Variable costs per case at GCP list prices (May 2026): Gemini 2.5 Flash approximately €0.003, Cloud Run CheXNet inference approximately €0.0001, Cloud Vision approximately €0.001, and storage plus egress

approximately €0.0001, for a total variable cost of roughly €0.005 to €0.01 per case. Fixed costs include salaries, clinical advisors, and regulatory consultants. One-off costs include the CE technical file, security audit, and pilot site setup.

Revenue Streams

Pay-per-case at €0.80 per case on the SMB tier, yielding approximately 98% gross margin on variable infrastructure cost. Monthly subscription at €499 for up to 1 000 cases with overage charges beyond that. Enterprise annual contracts on a volume-based basis including SLA, DICOM gateway access, and an on-prem deployment option. Research and academic pricing is discounted and paired with a data-use agreement.

4.1 Value Proposition Canvas

Customer Jobs

Rapidly triage incoming X-rays and prioritise urgent cases. Document findings for the patient file. Have confidence when no radiologist is on shift. Meet PNRR digitalisation KPIs.

Customer Pains

Four to twenty-four hour delays until a formal radiology read. Missing subtle findings such as pneumothorax or small effusions under fatigue. Inability to justify the capital cost of an enterprise AI appliance.

Pain Relievers

CheXNet probability scores and the Grad-CAM heatmap localise the abnormality visually. The LLM explanation drafts a finding line the clinician edits rather than writes from scratch.

Customer Gains

Same-shift triage decisions. The heatmap serves as a teaching tool for residents. Predictable per-case operational expenditure billable into the episode.

Gain Creators

Sub-60 second median end-to-end latency from upload to result. Multi-tenant SaaS requiring no IT installation. GDPR-resident infrastructure in the europe-west6 region.

5. Architectural Diagram

The diagram below shows the complete final-product architecture. The system is event-driven so a slow or failing AI service does not block uploads, and individual pipeline stages scale independently.

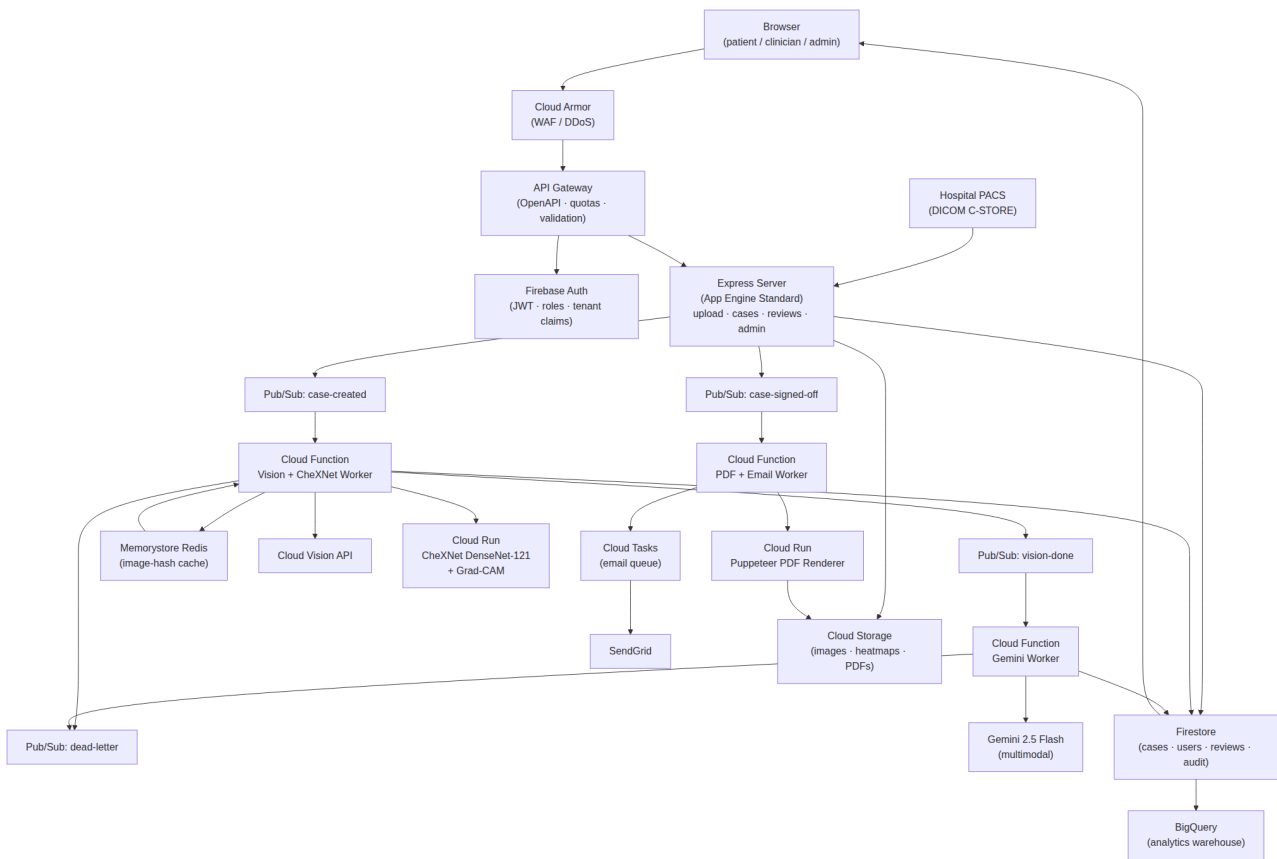


Figure 5.1 — MedTriageAI System Architecture

Component Responsibilities

Cloud Armor handles DDoS mitigation, IP allow and deny lists, and the OWASP managed rule set. The API Gateway provides TLS termination, OpenAPI-based request validation, and per-API-key quotas. Firebase Auth issues JWTs with role and tenantId custom claims. The DICOM gateway accepts DICOM C-STORE from hospital PACS, transcodes to JPEG, and injects into the upload pipeline.

The Express server exposes the HTTP API surface, validates uploads, publishes pipeline events, and handles clinician sign-off. Pub/Sub topics decouple pipeline stages with at-least-once delivery, retry, and dead-letter routing. The Vision and CheXNet worker runs Cloud Vision and CheXNet inference in parallel, consults the Redis cache first, and writes results to Firestore. The Gemini worker builds the LLM prompt from CheXNet scores and Vision labels and writes the final structured findings.

The PDF and email worker calls Puppeteer on sign-off to render the PDF, stores it in Cloud Storage, and enqueues the patient email via Cloud Tasks. MemoryStore Redis caches AI responses keyed by image SHA-256. Firestore is the source of truth for case state and provides the realtime channel back to the browser. Cloud Storage holds the original image, Grad-CAM heatmap PNG, and generated PDF, served via time-limited signed URLs. BigQuery and Looker Studio provide de-identified analytics. Secret Manager holds all credentials. Cloud Build and SonarCloud run the CI pipeline. Terraform provisions every GCP resource.

6. Use-Case Diagrams

6.1 Actors

Patient — uploads their own X-ray, views the AI-assisted result, and downloads the PDF report. Clinician (family doctor or ED physician) — reviews AI output, confirms or overrides findings, and signs off the report. Radiologist (optional reviewer) — provides an authoritative read on cases the clinician escalates. Administrator — manages user accounts, customer tenants, the audit trail, and billing. AI Pipeline — secondary actor representing the asynchronous worker chain. Hospital PACS — secondary actor representing DICOM-based image ingestion.

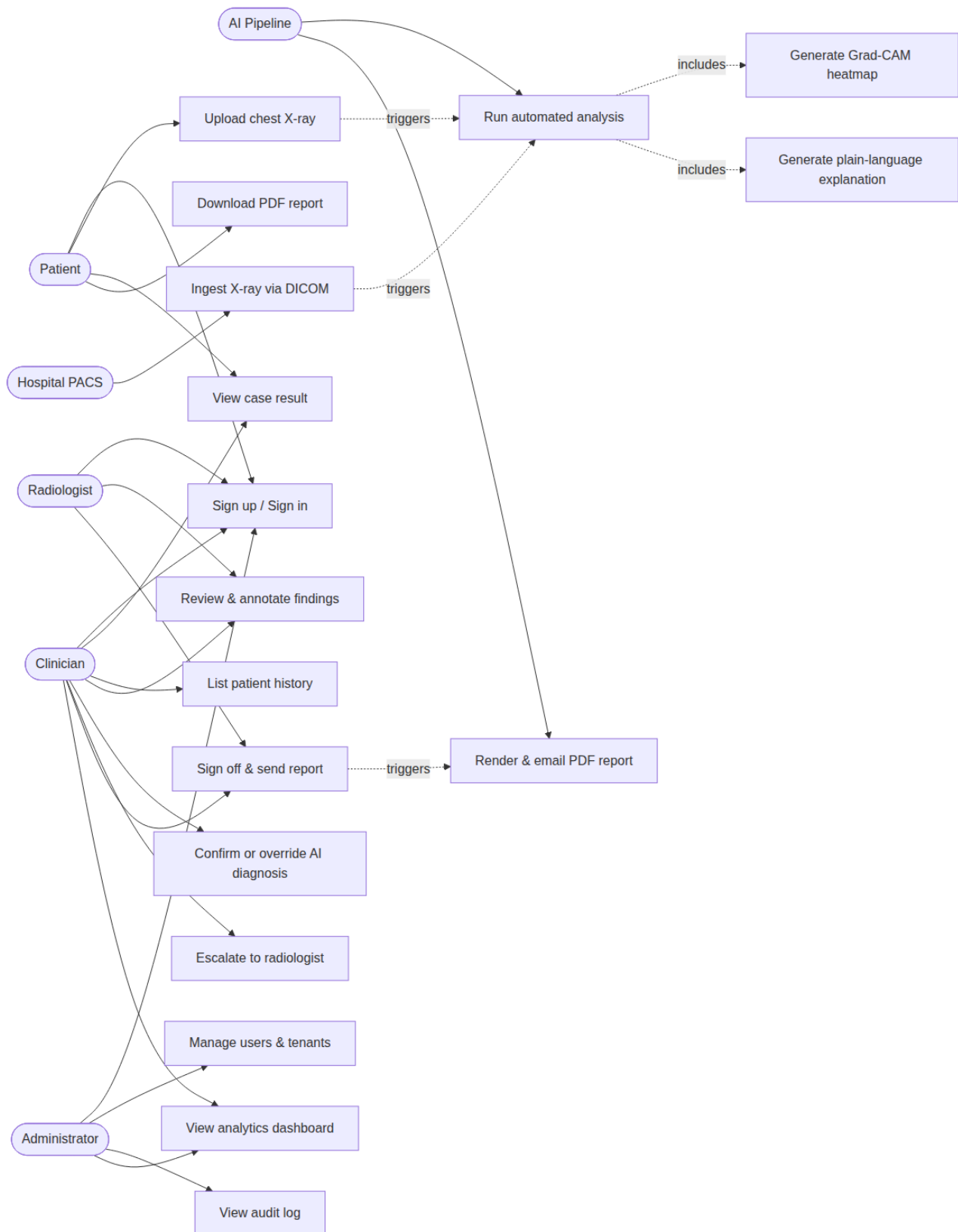


Figure 6.1 — System-Level Use-Case Diagram

6.2 Use-Case Detail: Upload Chest X-Ray (UC-02)

Primary actor: Patient, or Clinician on behalf of patient. Secondary actor: AI Pipeline. Preconditions: the actor is authenticated, the image is JPEG or PNG and does not exceed 10 MB, and consent for processing has been recorded.

Main flow: the actor selects a file in the browser and the browser POSTs to /api/upload as multipart. The server validates MIME type and file size, writes the image to Cloud Storage, creates the case document in Firestore, and publishes a case-created event to Pub/Sub, then returns the caseId. The browser subscribes to the Firestore case document via a realtime listener. The Vision and CheXNet worker receives the event, checks the Redis cache by image SHA-256, and on a cache miss runs Cloud Vision and CheXNet inference in parallel. It then updates the case document with scores and the heatmap and publishes vision-done. The Gemini worker picks up that event, builds its prompt, calls Gemini 2.5 Flash, and writes the final structured findings and explanation. The browser observes the status transition to complete and renders the result.

Alternate flow A: if CheXNet returns HTTP 503 due to a cold start, the worker retries up to three times with a 10 second back-off before failing the message. Alternate flow B: if Gemini returns malformed JSON, the raw text is stored in the explanation field and the classification falls back to Other. Exception: any unhandled exception is captured, the pipeline message is routed to the dead-letter topic, and the case is marked with status error.

6.3 Use-Case Detail: Clinician Review and Sign-Off (UC-09)

Primary actor: Clinician. Preconditions: the case exists with status complete and the clinician role has access to the tenant.

Main flow: the clinician opens the case. The UI displays the original image, the Grad-CAM heatmap, the CheXNet probability scores, and the Gemini explanation. The clinician edits the findings text and adjusts severity if needed, then clicks Sign off. The system creates a review subdocument with the clinician identity, timestamp, and a hash of the finalised content, then publishes case-signed-off. The PDF worker renders the report via Puppeteer, stores it in Cloud Storage, and generates a signed URL. The email worker enqueues a SendGrid send to the patient with the report link. Alternate flow: if the clinician selects Escalate to radiologist, the case status moves to awaiting_radiologist.

6.4 Use-Case Detail: DICOM Ingestion (UC-02D)

Primary actor: Hospital PACS. Preconditions: the customer has configured a DICOM AE title for MedTriageAI and mTLS is established.

Main flow: the PACS issues a DICOM C-STORE to the DICOM gateway. The gateway extracts the pixel data, anonymises PHI tags according to tenant configuration, and transcodes the image to JPEG. It then calls the internal /api/upload endpoint on behalf of the configured tenant and the regular UC-02 pipeline runs. Once the case reaches complete status, the result is posted back to the customer via a configured FHIR DiagnosticReport webhook.

7. APIs and Functionality Flows

The API contract is defined in OpenAPI 3.0 and published on SwaggerHub at <https://app.swaggerhub.com/apis/MedTriageAI/medtriage-api/1.0.0>. The full YAML is in the repository at [docs/openapi.yaml](#). All endpoints require TLS. Authentication uses session JWTs issued by Firebase Auth except where noted.

7.1 Endpoint Overview

POST `/api/auth/session`, authenticated with a Firebase ID token, exchanges the token for a backend session JWT used on all subsequent requests.

POST `/api/upload`, multipart/form-data with session JWT, accepts a chest X-ray image, writes it to Cloud Storage, creates a case document, publishes the pipeline event, and returns the `caseId` as a 202 Accepted response.

POST `/api/dicom`, DICOMweb STOW-RS with mTLS client certificate, is the DICOM ingestion endpoint for hospital PACS systems that internally calls the upload pipeline after transcoding.

GET `/api/cases`, session JWT required, returns a paginated list of cases for the authenticated user or tenant.

GET `/api/cases/{caseId}`, session JWT required, returns the full case document including CheXNet scores, heatmap reference, Gemini findings, severity, and status.

POST `/api/cases/{caseId}/review`, clinician role required, accepts edited findings text, severity, and an optional sign-off flag, creates a review subdocument and, if signed off, triggers the PDF and email pipeline.

POST `/api/cases/{caseId}/escalate`, clinician role required, moves case status to `awaiting_radiologist` and routes it to the radiologist review queue.

GET `/api/cases/{caseId}/report.pdf`, session JWT required, returns a redirect to the Cloud Storage signed URL for the generated PDF report.

GET, POST, DELETE `/api/admin/users`, admin role required, manages user accounts and role assignments within the tenant.

GET `/api/admin/audit`, admin role required, queries the structured audit log stored in Firestore and exported to BigQuery.

GET `/api/admin/metrics`, admin role required, returns tenant-level analytics including case volume and severity distribution.

GET `/api/health`, public, liveness probe used by App Engine and external uptime monitors.

GET `/api/metrics`, internal, Prometheus-format metrics scraped by Cloud Monitoring.

7.2 Key OpenAPI Schemas

The Case schema includes: `caseId` as a UUID, `tenantId`, `status` as an enum (`pending`, `vision_processing`, `vision_done`, `analyzing`, `complete`, `awaiting_radiologist`, `signed_off`, `error`), `timestamp`, `imageUrl`, `heatmap` as a base64-encoded PNG string, `visionLabels` as an array of description and score pairs, `chexnetScores` as an object mapping pathology name to probability between 0 and 1, `chexnetTopFindings` as an array of name and score objects, `classification`, `severity` as an enum (`normal`, `mild`, `moderate`, `severe`, `unknown`), `findings`, `explanation`, and a nullable error field.

The Review schema includes: findings as a string, severity as an enum (normal, mild, moderate, severe), and signOff as a boolean.

7.3 Functionality Flow: Upload to AI Pipeline to Result

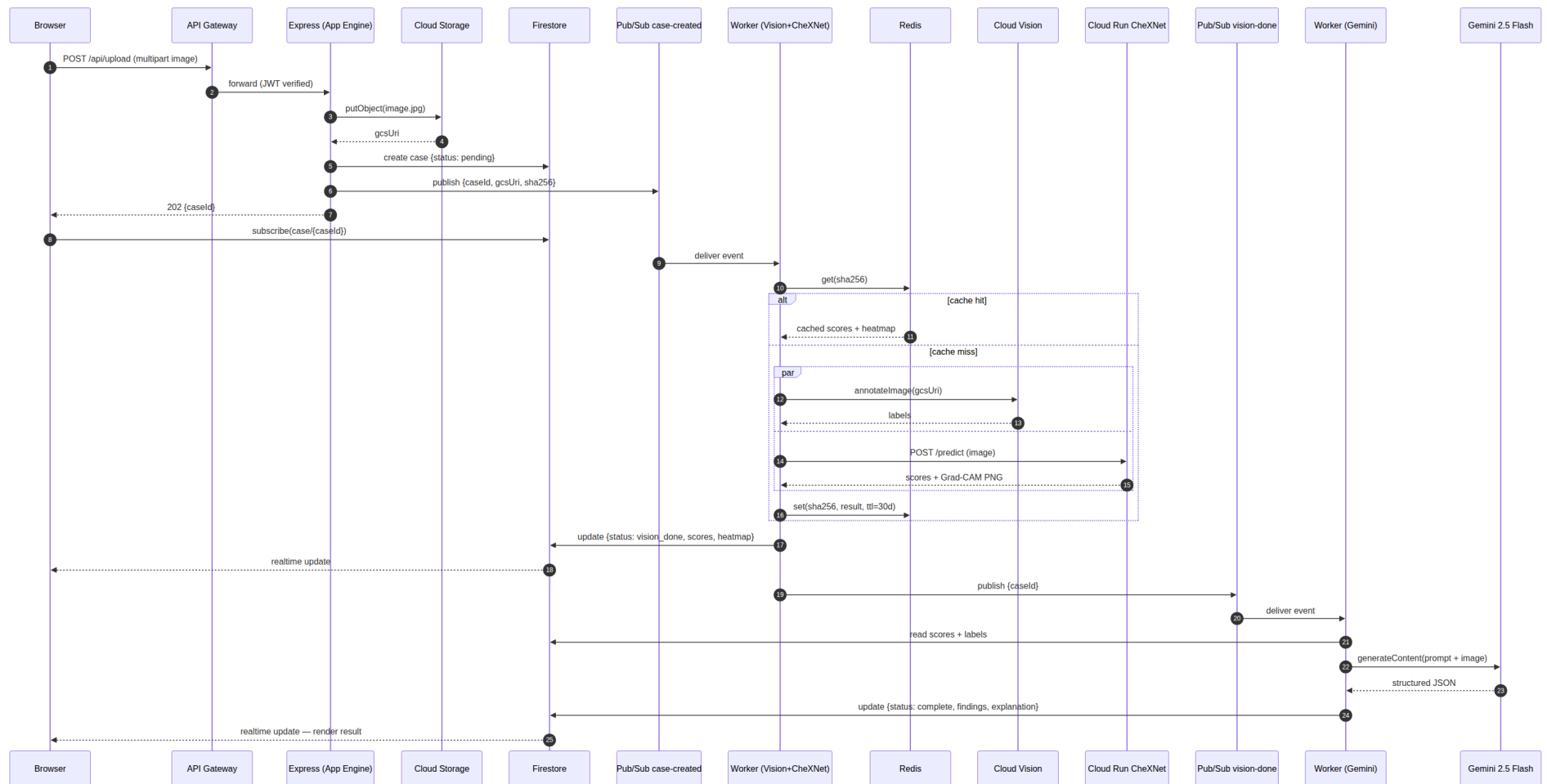


Figure 7.1 — Upload and AI Pipeline Sequence Diagram

7.4 Functionality Flow: Clinician Sign-Off and Report Delivery



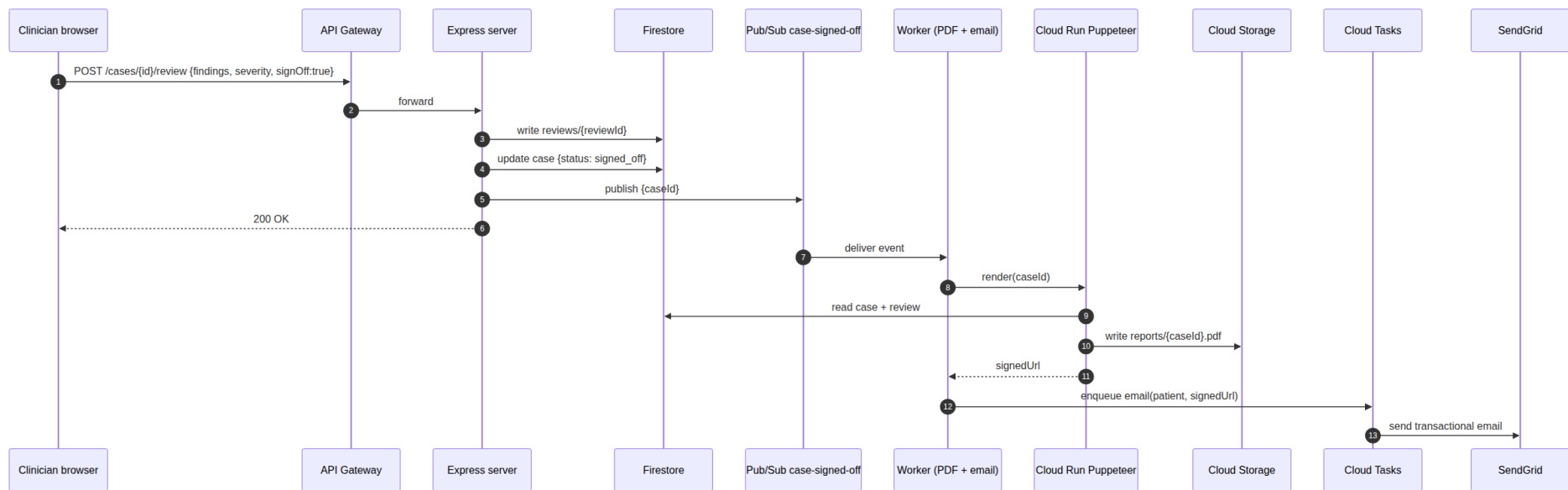


Figure 7.2 — Clinician Sign-Off and PDF Delivery Sequence Diagram

7.5 Error and Retry Semantics

Worker failures at any pipeline stage are retried by Pub/Sub with exponential back-off. After five delivery attempts the message is routed to the dead-letter topic and the case is marked with status error. CheXNet cold-start errors returning HTTP 503 are retried in-worker up to three times with a 10 second back-off before the pipeline message is failed. Gemini JSON parse failures are not retried: the raw text is stored in the explanation field and the classification falls back to Other. Every pipeline message carries the caseId and workers check the current case status before writing, making duplicate deliveries safe. Outbound webhook and email delivery uses Cloud Tasks which retries with exponential back-off; persistent failures surface to a webhook dead-letter queue topic.